

Programming for Economist

PS 2

Class 9

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Plan for today

1. Hints for PS2
2. Work on PS2
3. If you want I can go through some of the questions on the board. Just let me know:)

1. Hints for PS2

- Q1

- Check if x_1 and $x_2 > 0$ else return zero or $-\infty$, to prevent the optimisers from choosing negative values of x_1 and x_2

Use:

if 'fill in':

 return 'fil in'

else:

 return 'fil in'

- Q5

Newton's method: $x_k = x - \frac{f(x)}{df(x)}$ where x_k is the next guess and x is the current

- Q6

Define the objective function as a lambda function with 1 input:

`obj = lambda x1: -value_of_choice_ces(x1,,alpha,beta,p1,p2,I)`

1. Hints for PS2

- Q7

To time you need to 'import time'. Then use

`'t0 = time.perf_counter()'` and `'t1 = time.perf_counter()'`

Before and after the solver. Subtract the two to get the runtime.

When defining the objective function have `x1` be the index of the input. This is due to the way minimize work. Use

```
def obj(x):
```

```
    x1 = x[0]
```

```
    'fill the rest'
```

- Q8

Compute the required variables using an optimiser. Use Jeppe's plotting code. As always make sure you understand code you didn't write.